

Part 1: Dynamic Tables – Core Practice

1. Create a source table `orders_src` with:

- `order_id`
- `order_date`
- `region`
- `product`
- `amount`

```
1 create database management;
2 CREATE OR REPLACE TABLE orders_src (
3     order_id INT,
4     order_date DATE,
5     region STRING,
6     product STRING,
7     amount NUMBER(10,2)
8 );
```

Results | Chart

#	status	Query Details
1	Table ORDERS_SRC successfully created.	Query duration 185ms

2. Insert at least 8 records covering multiple regions and products.

```
10 INSERT INTO orders_src VALUES
11 (1, '2025-01-01', 'North', 'Laptop', 50000),
12 (2, '2025-01-02', 'South', 'Mobile', 20000),
13 (3, '2025-01-03', 'East', 'Tablet', 15000),
14 (4, '2025-01-04', 'West', 'Laptop', 52000),
15 (5, '2025-01-05', 'North', 'Mobile', 22000),
16 (6, '2025-01-06', 'South', 'Tablet', 18000),
17 (7, '2025-01-07', 'East', 'Laptop', 48000),
18 (8, '2025-01-08', 'West', 'Mobile', 21000);
```

Results | Chart

#	number of rows inserted	Query Details
	8	Query duration 817ms

3. Create a Dynamic Table `orders_region_summary_dt` that shows:

- `region`
- `total_orders`
- `Total_sales` Set `TARGET_LAG = 5 MINUTE`.

```
20 CREATE OR REPLACE DYNAMIC TABLE orders_region_summary_dt
21 TARGET_LAG = '5 MINUTE'
22 WAREHOUSE = COMPUTE_WH
23 AS
24 SELECT
25     region,
26     COUNT(order_id) AS total_orders,
27     SUM(amount) AS total_sales
28 FROM orders_src
29 GROUP BY region;
```

Results | Chart

#	status	Query Details
1	Dynamic table ORDERS_REGION_SUMMARY_DT successfully created.	Query duration 1.7s

Initial Check :

```
30
31 | SELECT * FROM orders_region_summary_dt;
32
```

	REGION	# TOTAL_ORDERS	# TOTAL_SALES
1	East	2	63000.00
2	North	2	72000.00
3	South	2	38000.00
4	West	2	73000.00

Query Details
Query duration 178ms
Rows 4
Query ID 01c1bfb3-0001-7de7-00...
[Show more](#)

4. Insert new records into orders_src and verify that the Dynamic Table updates automatically.

```
34 | INSERT INTO orders_src VALUES
35 | (9, '2025-01-09', 'North', 'Tablet', 17000),
36 | (10, '2025-01-10', 'South', 'Laptop', 55000);
37
```

	# number of rows inserted
1	2

Query Details
Query duration 259ms

Wait up to 5 minutes, then:

```
38 | SELECT * FROM orders_region_summary_dt;
39
```

	REGION	# TOTAL_ORDERS	# TOTAL_SALES
1	North	3	89000.00
2	South	3	93000.00
3	East	2	63000.00
4	West	2	73000.00

Query Details
Query duration 133ms
Rows 4
Query ID 01c1bfb3-0001-7de7-00...
[Show more](#)

Final result :

- North total_orders increases
- South total_orders & total_sales increase
- No manual refresh needed → Dynamic Table auto-updates

5. Update the amount for an existing order and observe the Dynamic Table behavior.

```
40 | UPDATE orders_src
41 | SET amount = 60000
42 | WHERE order_id = 1;
43
```

	# number of rows updated	# number of multi-joined rows updated
1	1	0

Query Details
Query duration 288ms

After TARGET_LAG:

```
38 | SELECT * FROM orders_region_summary_dt;
```

	REGION	TOTAL_ORDERS	TOTAL_SALES
1	North	3	89000.00
2	South	3	93000.00
3	East	2	63000.00
4	West	2	73000.00

Query Details

Query duration 58ms

Rows 4

Query ID 01c1bfbc-0001-7de7-00...

Show more

Final result :

- North total_sales increases
- Order count remains unchanged
- Dynamic Table reflects the update automatically

6. Delete one record from orders_src.

Re-query the Dynamic Table and note the result.

```
44 | DELETE FROM orders_src
```

```
45 | WHERE order_id = 2;
```

	number of rows deleted
1	1

Query Details

Query duration 264ms

Re-query after lag :

```
38 | SELECT * FROM orders_region_summary_dt;
```

	REGION	TOTAL_ORDERS	TOTAL_SALES
1	East	2	63000.00
2	West	2	73000.00
3	North	3	99000.00
4	South	3	93000.00

Query Details

Query duration 131ms

Rows 4

Query ID 01c1bfbe-0001-7de7-00...

Show more

Final result :

- South total_orders decreases by 1
- South total_sales decreases
- Dynamic Table reflects delete automatically

7. Suspend the warehouse used by the Dynamic Table.

Query the Dynamic Table and observe what happens.

47ALTER WAREHOUSE COMPUTE_WH SUSPEND;

Results

Chart

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	status
1	Statement executed successfully.

Query Details

...

Query duration68ms

Now query:

38

SELECT * FROM orders_region_summary_dt;

39

Results

Chart

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	REGION	# TOTAL_ORDERS	# TOTAL_SALES
1	North	3	99000.00
2	East	2	63000.00
3	West	2	73000.00
4	South	2	73000.00

Query Details

...

Query duration

574ms

Rows

4

Query ID

01c1bfc1-0001-7de7-00...

Show more

▼

Final result :

- Query **fails or returns stale data**
- Dynamic Table **cannot refresh** without an active warehouse
- Once warehouse resumes, it refreshes automatically

49

ALTER WAREHOUSE COMPUTE_WH RESUME;

Results

Chart

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	status
1	Statement executed successfully.

Query Details

⋮

Query duration54ms

Part 2: Dynamic Table Dependency Chain

- Create a Dynamic Table orders_clean_dt that:
 - Filters out records where amount <= 0.

```
51 CREATE OR REPLACE DYNAMIC TABLE orders_clean_dt
52 TARGET_LAG = '1 MINUTE'
53 WAREHOUSE = compute_wh
54 AS
55 SELECT
56     order_id,
57     order_date,
58     region,
59     product,
60     amount
61 FROM orders_src
62 WHERE amount > 0;
```

Results	Chart	
A status	Query Details ... Query duration 833ms	
1 Dynamic table ORDERS_CLEAN_DT successfully created.		

- Create another Dynamic Table orders_product_kpi_dt that:
 - Depends on orders_clean_dt
 - Shows total_sales and total_orders by product.

This Dynamic Table **depends on** `orders_clean_dt`, not the base table.

```
54 CREATE OR REPLACE DYNAMIC TABLE orders_product_kpi_dt
55 TARGET_LAG = '1 MINUTE'
56 WAREHOUSE = compute_wh
57 AS
58 SELECT
59     product,
60     COUNT(order_id) AS total_orders,
61     SUM(amount) AS total_sales
62 FROM orders_clean_dt
63 GROUP BY product;
```

→ Results Chart

status
Dynamic table ORDERS_PRODUCT_KPI_DT successfully created.

Query Details ...

Query duration 1.2s

Snowflake automatically understands the dependency chain:



10. Insert invalid and valid records into `orders_src`.
Verify how both Dynamic Tables behave.

```
75 INSERT INTO orders_src VALUES
76 (201, '2024-01-10', 'East', 'Laptop', 1200),
77 (202, '2024-01-11', 'West', 'Mobile', -500), -- X Invalid
78 (203, '2024-01-12', 'South', 'Tablet', 0), -- X Invalid
79 (204, '2024-01-13', 'North', 'Laptop', 1500), -- ✓ Valid
80 (205, '2024-01-14', 'East', 'Mobile', 800); -- ✓ Valid
81
```

→ Results Chart

#	number of rows inserted	:
1	5	

Query Details ...

Query duration 257ms

```

82 | SELECT * FROM orders_clean_dt;
83 |

```

	# ORDER_ID	ORDER_DATE	REGION	PRODUCT	# AMOUNT
1	1	2025-01-01	North	Laptop	60000.00
2	3	2025-01-03	East	Tablet	15000.00
3	4	2025-01-04	West	Laptop	52000.00
4	5	2025-01-05	North	Mobile	22000.00
5	6	2025-01-06	South	Tablet	18000.00
6	7	2025-01-07	East	Laptop	48000.00
7	8	2025-01-08	West	Mobile	21000.00
8	9	2025-01-09	North	Tablet	17000.00

Query Details

Query duration 59ms

Rows 15

Query ID 01c1bfdb-0001-7de7-00...

Show more

ORDER_ID #

Only records with **amount > 0** appear.

```

84 | SELECT * FROM orders_product_kpi_dt;
85 |

```

	PRODUCT	# TOTAL_ORDERS	# TOTAL_SALES
1	Mobile	4	44600.00
2	Laptop	8	221200.00
3	Tablet	3	50000.00

Query Details

Query duration 143ms

Rows 3

Query ID 01c1bfdb-0001-7de7-00...

KPIs calculated **only from clean data**.

11. Change data in the base table and confirm refresh order without manual scheduling.

```

86 | UPDATE orders_src
87 | SET amount = 2000
88 | WHERE order_id = 201;

```

	# number of rows updated	# number of multi-joined rows updated
1	2	0

Query Details

Query duration 287ms

- **orders_clean_dt** refreshes first
- **orders_product_kpi_dt** refreshes next
- No manual task, no cron, no trigger

Part 3: SCD Type 1 Using Dynamic Tables

12. Create a source table customer_src with:

- customer_id
- customer_name
- city
- updated_at

```

90 CREATE OR REPLACE TABLE customer_src (
91     customer_id INT,
92     customer_name STRING,
93     city STRING,
94     updated_at TIMESTAMP
95 );

```

Results | Chart

status

Table CUSTOMER_SRC successfully created.

Query Details

Query duration 145ms

13. Insert multiple records per customer with different timestamps.

```

98 INSERT INTO customer_src VALUES
99 (1, 'Ravi', 'Chennai', '2024-01-01 10:00:00'),
100 (1, 'Ravi', 'Bangalore', '2024-01-05 12:00:00'),
101 (2, 'Anita', 'Delhi', '2024-01-02 09:30:00'),
102 (2, 'Anita', 'Mumbai', '2024-01-06 15:45:00'),
103 (3, 'John', 'Hyderabad', '2024-01-03 11:15:00');
104
105

```

Results | Chart

number of rows inserted

5

Query Details

Query duration 294ms

- Same customer appears multiple times
- Each row represents a **new version**

14. Create a Dynamic Table customer_dim_scd1_dt that keeps only the latest record per customer.

Answer :

- Keeps **only the latest record per customer**
- Older versions are **automatically removed**

```

106 CREATE OR REPLACE DYNAMIC TABLE customer_dim_scd1_dt
107     TARGET_LAG = '1 MINUTE'
108     WAREHOUSE = compute_wh
109     AS
110     SELECT
111         customer_id,
112         customer_name,
113         city,
114         updated_at
115     FROM customer_src
116     QUALIFY ROW_NUMBER() OVER (
117         PARTITION BY customer_id
118         ORDER BY updated_at DESC
119     ) = 1;

```

Results | Chart

status

Dynamic table CUSTOMER_DIM_SCD1_DT successfully created.

Query Details

Query duration 812ms

Why this works (SCD Type 1 logic)

- **ROW_NUMBER()** ranks rows per customer
- Latest timestamp gets rank = 1
- Dynamic Table always shows current version only

15. Insert a new version of an existing customer and verify that old data disappears.

```
122 INSERT INTO customer_src VALUES
123 (1, 'Ravi', 'Pune', '2024-01-10 09:00:00');
124
125
```

Results		Chart		Query Details	
# number of rows inserted		1		Query duration 399ms	

Check Dynamic Table

```
126 SELECT *
127 FROM customer_dim_scd1_dt
128 WHERE customer_id = 1;
129
```

Results		Chart		Query Details	
# CUSTOMER_ID	A CUSTOMER_NAME	A CITY	UPDATED_AT	Query duration 145ms	
1	Ravi	Pune	2024-01-10 09:00:00.000		

- Old cities (Chennai, Bangalore) are gone
- Only **latest data exists**

Final result :

- SCD Type 1 with Dynamic Tables means “keep only the latest customer details and forget the past.”
- Snowflake does this automatically whenever new data arrives.

Part 4: Stored Procedures – ETL Practice

16. Create a target table orders_audit.

```
132 CREATE OR REPLACE TABLE orders_audit (
133     audit_id      INT AUTOINCREMENT,
134     load_time     TIMESTAMP,
135     record_count  INT
136 );
```

Results		Chart		Query Details	
status		1		Query duration 143ms	
Table ORDERS_AUDIT successfully created.				Rows 1	

17. Write a Stored Procedure that:

- Inserts rows into orders_audit
- Captures load timestamp
- Captures record count


```

139 CREATE OR REPLACE PROCEDURE sp_load_orders_audit()
140 RETURNS STRING
141 LANGUAGE SQL
142 AS
143 $$
144 DECLARE
145     v_count INT;
146 BEGIN
147     -- Count records from source
148     SELECT COUNT(*) INTO :v_count FROM orders_src;
149
150     -- Insert audit record
151     INSERT INTO orders_audit (load_time, record_count)
152     VALUES (CURRENT_TIMESTAMP(), v_count);
153
154     RETURN 'Audit load successful';
155 END;
156 $$;

```

Results Chart

status

Function SP_LOAD_ORDERS_AUDIT successfully created.

Query Details

Query duration 53ms

- Captures **load timestamp**
- Captures **record count**

18. Execute the Stored Procedure manually.

```

137 CREATE OR REPLACE PROCEDURE sp_load_orders_audit()
138 RETURNS STRING
139 LANGUAGE SQL
140 EXECUTE AS OWNER
141 AS
142 $$
143 DECLARE
144     v_count INTEGER;
145 BEGIN
146     -- Get record count
147     v_count := (SELECT COUNT(*) FROM orders_src);
148
149     -- Insert audit record
150     INSERT INTO orders_audit (load_ts, record_count)
151     VALUES (CURRENT_TIMESTAMP(), v_count);
152
153     RETURN 'SUCCESS: Orders audit Loaded. Records = ' || v_count;
154
155 EXCEPTION
156 WHEN OTHER THEN
157     RETURN 'ERROR: ' || SQLERRM;
158 END;
159 $$;

```

Results Chart

status

Function SP_LOAD_ORDERS_AUDIT successfully created.

Query Details

Query duration 62ms

```

169 SHOW PROCEDURES LIKE 'SP_LOAD_ORDERS_AUDIT';
170 CALL sp_load_orders_audit();
171
172

```

Results Chart

	created_on	name	schema_name	is_builtin	is_aggregate	is_ansi	min_num_arguments	max_num_arguments
1	2026-01-15 07:04:43	SP_LOAD_ORDERS_AUDIT	PUBLIC	N	N	N	0	

Query Details

Query duration 29ms

19. Schedule the Stored Procedure using a Task.

Create task : Runs **every 1 hour**

```

182 CREATE OR REPLACE TASK task_load_orders_audit
183 WAREHOUSE = compute_wh
184 SCHEDULE = 'USING CRON @ * / 1 * * * UTC'
185 AS
186 CALL sp_load_orders_audit();
187

```

Results Chart

status

Task TASK_LOAD_ORDERS_AUDIT successfully created.

Query Details

Query duration 61ms

Enable task :

188ALTER TASK task_load_orders_audit RESUME;
189

ResultsChart

status

1Statement executed successfully.

Query Details

Query duration82ms

Verify task :

190

SHOW TASKS;

191

Results

Chart

created_on

name

id

database_name

schema_name

owner

comment

warehouse

schedule

1

2026-01-15 07:11:32

TASK_LOAD_ORDERS_

01c1c00f-78e8-a6c7-

MANAGEMENT

PUBLIC

ACCOUNTADMIN

COMPUTE_WH

USING CRON 0 */1 *

Query Details

Query duration

31ms

Part 5: Stored Procedures vs Dynamic Tables

Feature	Stored Procedure	Dynamic Table
UPDATE / DELETE	✔ Yes	✘ No
Conditional logic	✔ Yes	✘ Limited
Modifies source table	✔ Yes	✘ No
Auto refresh	✘ (Task needed)	✔ Built-in
Best for	ETL, DML, business rules	Reporting, transformations

21. Use a Stored Procedure to to:
- Update a flag column in a table conditionally.

1) Create source table

```
192 CREATE OR REPLACE TABLE orders_flag (
193     order_id INT,
194     amount NUMBER,
195     high_value_flag STRING
196 );
197
```

Results	Chart	Query Details
---------	-------	---------------

status	Query duration	135ms
--------	----------------	-------

Table ORDERS_FLAG successfully created.		
---	--	--

2) Insert sample data

```
199 | INSERT INTO orders_flag VALUES
200 | (1, 500, NULL);
201 | (2, 1500, NULL);
202 | (3, 3000, NULL);
203 |
204 |
```

Results

Chart

number of rows inserted

1

Query Details

...

Query duration

314ms

3) Stored Procedure to update flag conditionally

```

205 CREATE OR REPLACE PROCEDURE sp_update_high_value_flag()
206 RETURNS STRING
207 LANGUAGE SQL
208 AS
209 $$
210 BEGIN
211     UPDATE orders_flag
212     SET high_value_flag =
213     CASE
214         WHEN amount > 1000 THEN 'Y'
215         ELSE 'N'
216     END;
217     RETURN 'Flag updated successfully';
218 END;
219 $$;
220
221
222

```

Results | Chart

status

Function SP_UPDATE_HIGH_VALUE_FLAG successfully created.

Query Details

Query duration 45ms

4 Execute procedure

```

224 CALL sp_update_high_value_flag();
225
226

```

Results | Chart

SP_UPDATE_HIGH_VALUE_FLAG

Flag updated successfully

Query Details

Query duration 462ms

5 Verify

```

228 SELECT * FROM orders_flag;
229
230

```

Results | Chart

#	ORDER_ID	#	AMOUNT		HIGH_VALUE_FLAG
1	1		500		N
2	2		1500		Y
3	3		3000		Y

Query Details

Query duration 136ms

Rows 3

Query ID 01c1c046-0001-7de7-0...

22. Try implementing the same logic using a Dynamic Table.
Note the limitation.

```

230 CREATE OR REPLACE DYNAMIC TABLE orders_flag_dt
231 TARGET_LAG = '1 minute'
232 WAREHOUSE = compute_wh
233 AS
234 SELECT
235     order_id,
236     amount,
237     CASE
238         WHEN amount > 1000 THEN 'Y'
239         ELSE 'N'
240     END AS high_value_flag
241 FROM orders_flag;
242
243

```

Results | Chart

status

Dynamic table ORDERS_FLAG_DT successfully created.

Query Details

Query duration 91ms

This creates a table .But does not update orders_flag.

Why Dynamic Table FAILS for this requirement

- Dynamic Tables are READ-ONLY derived objects

They are :

- Cannot UPDATE existing tables
- Cannot modify source data
- Cannot run procedural logic

- Cannot execute conditional DML

They only:

- SELECT
- Transform
- Auto-refresh results

Part 6: Dashboard-Ready Queries

23. Write a query on orders_region_summary_dt for:

- Total sales by region (descending).

```
244 SELECT
245     region,
246     SUM(total_sales) AS total_sales
247 FROM orders_region_summary_dt
248 GROUP BY region
249 ORDER BY total_sales DESC;
250
```

REGION	TOTAL_SALES
North	102000.00
South	73000.00
West	72000.00
East	68600.00

Query Details

Query duration 669ms

Rows 4

Query ID 01c1c054-0001-7de7-0...

24. Write a query on orders_product_kpi_dt for:

- Top 3 products by sales.

```
253 SELECT
254     product,
255     SUM(total_sales) AS total_sales
256 FROM orders_product_kpi_dt
257 GROUP BY product
258 ORDER BY total_sales DESC
259 LIMIT 3;
260
```

PRODUCT	TOTAL_SALES
Laptop	222000.00
Tablet	50000.00
Mobile	44600.00

Query Details

Query duration 259ms

Rows 3

25. Filter dashboard data for:

- Region = 'East'

```
266 SELECT *
267 FROM orders_src
268 WHERE region = 'East';
269
```

ORDER_ID	ORDER_DATE	REGION	PRODUCT	AMOUNT
3	2025-01-03	East	Tablet	15000.00
7	2025-01-07	East	Laptop	48000.00
205	2024-01-14	East	Mobile	800.00
201	2024-01-10	East	Laptop	2000.00

Query Details

Query duration 26ms

Rows 5

Query ID 01c1c064-0001-7de7-0...

- Product = 'Laptop'

```
271 SELECT *
272 FROM orders_product_kpi_dt
273 WHERE product = 'Laptop';
274
```

PRODUCT	TOTAL_ORDERS	TOTAL_SALES
Laptop	8	222000.00

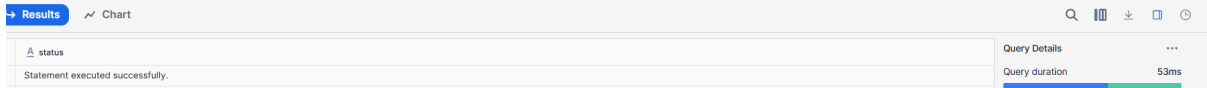
Query Details

Query duration 48ms

Part 7: Monitoring & Operations

27. Change TARGET_LAG to a higher value and observe refresh behavior.

```
320
321 ALTER DYNAMIC TABLE orders_region_summary_dt
322 SET TARGET_LAG = '30 minutes';
323
```



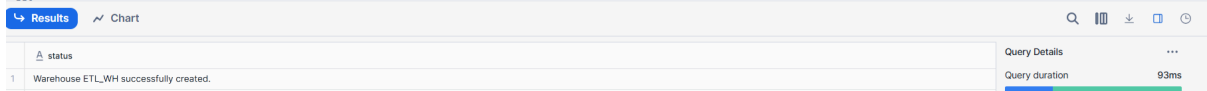
The screenshot shows a query execution interface. The top bar has a 'Results' tab and a 'Chart' tab. Below the tabs, there's a status bar indicating 'Statement executed successfully.' To the right, a 'Query Details' panel shows 'Query duration' as 53ms.

- Refresh happens **less frequently**
- **Lower compute cost**
- Data may be **slightly less fresh**
- Good for **dashboards**, not real-time analytics

28. Separate compute by assigning:

- One warehouse for ETL

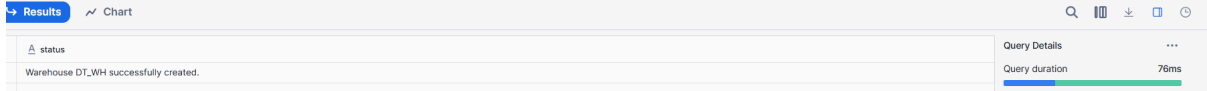
```
324
325 CREATE OR REPLACE WAREHOUSE etl_wh
326 WAREHOUSE_SIZE = MEDIUM
327 AUTO_SUSPEND = 60
328 AUTO_RESUME = TRUE;
329
```



The screenshot shows a query execution interface. The top bar has a 'Results' tab and a 'Chart' tab. Below the tabs, there's a status bar indicating 'Warehouse ETL_WH successfully created.' To the right, a 'Query Details' panel shows 'Query duration' as 93ms.

- One warehouse for Dynamic Tables

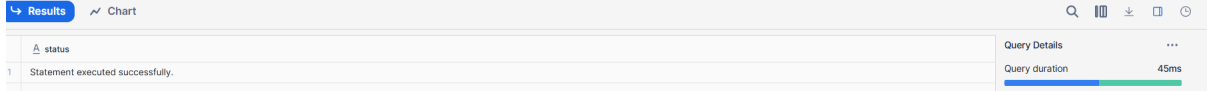
```
330
331 CREATE OR REPLACE WAREHOUSE dt_wh
332 WAREHOUSE_SIZE = SMALL
333 AUTO_SUSPEND = 60
334 AUTO_RESUME = TRUE;
335
```



The screenshot shows a query execution interface. The top bar has a 'Results' tab and a 'Chart' tab. Below the tabs, there's a status bar indicating 'Warehouse DT_WH successfully created.' To the right, a 'Query Details' panel shows 'Query duration' as 76ms.

Assign warehouse to Dynamic Table :

```
334
335 ALTER DYNAMIC TABLE orders_region_summary_dt
336 SET WAREHOUSE = dt_wh;
337
```



The screenshot shows a query execution interface. The top bar has a 'Results' tab and a 'Chart' tab. Below the tabs, there's a status bar indicating 'Statement executed successfully.' To the right, a 'Query Details' panel shows 'Query duration' as 45ms.

✓ Why this separation is important

Purpose	Benefit
ETL warehouse	Heavy transformations won't affect dashboards
Dynamic Table warehouse	Predictable refresh & stable performance
Separate cost tracking	Easy billing & optimization

Part 8: Failure & Recovery Practice

29. Introduce invalid SQL in a Dynamic Table definition and observe behavior.

1) Introduce Invalid SQL in a Dynamic Table (Example: Create a Dynamic Table (working version))

```
378 CREATE OR REPLACE DYNAMIC TABLE orders_summary_dt
379   TARGET_LAG = '1 MINUTE'
380   WAREHOUSE = compute_wh
381   AS
382   SELECT
383     region,
384     COUNT(*) AS total_orders,
385     SUM(amount) AS total_sales
386   FROM orders_src
387   GROUP BY region;
```

Results | Chart

status
1 Dynamic table ORDERS_SUMMARY_DT successfully created.

Query Details

Query duration 1.7s

✗ Introduce an Error (Invalid Column)

Modify the Dynamic Table with an invalid column:

```
390 CREATE OR REPLACE DYNAMIC TABLE orders_summary_dt
391   TARGET_LAG = '1 MINUTE'
392   WAREHOUSE = compute_wh
393   AS
394   SELECT
395     region,
396     COUNT(*) AS total_orders,
397     SUM(invalid_amount) AS total_sales -- ✗ column does not exist
398   FROM orders_src
399   GROUP BY region;
```

Results | Chart

Error: invalid identifier 'INVALID_AMOUNT' (line 394)

Query Details

Query duration 37ms

Rows 0

Query ID 01c1c324-0001-7eb2-0...

Observe Behavior:

```
402 DESCRIBE DYNAMIC TABLE orders_summary_dt;
```

Results | Chart

name	type	kind	null?	default	primary key	unique key	check	expression	comment	policy name
REGION	VARCHAR(16777216)	COLUMN	Y	null	N	N	null	null	null	null
TOTAL_ORDER	NUMBER(18,0)	COLUMN	Y	null	N	N	null	null	null	null
TOTAL_SALES	NUMBER(22,2)	COLUMN	Y	null	N	N	null	null	null	null

Query Details

Query duration 69ms

Rows 3

Query ID 01c1c324-0001-7e8f-00...

What happens:

- Dynamic Table **enters FAILED state**
- Data **does NOT refresh**
- Error message shows **invalid column**
- Snowflake does **not drop** the Dynamic Table
- Last successful data (if any) remains available

30. Fix the issue and confirm auto-recovery.

```
405 CREATE OR REPLACE DYNAMIC TABLE orders_summary_dt
406 TARGET_LAG = '1 MINUTE'
407 WAREHOUSE = compute_wh
408 AS
409 SELECT
410     region,
411     COUNT(*) AS total_orders,
412     SUM(amount) AS total_sales -- [X] fixed
413 FROM orders_src
414 GROUP BY region;
415
```

Results | Chart

status
Dynamic table ORDERS_SUMMARY_DT successfully created.

Query Details

Query duration 1.4s

Auto-Recovery Behavior :

- Dynamic Table **automatically resumes refresh**
- Status changes from **FAILED** → **ACTIVE**
- No manual restart required

31. Stop the Task running the Stored Procedure and restart it.

```
416 CREATE OR REPLACE TASK orders_audit_task
417 WAREHOUSE = compute_wh
418 SCHEDULE = '5 MINUTE'
419 AS
420 CALL load_orders_audit();
421
422
423
```

Results | Chart

status
Task ORDERS_AUDIT_TASK successfully created.

Query Details

Query duration 121ms

Suspend the Task

```
424 ALTER TASK orders_audit_task SUSPEND;
425
```

Results | Chart

status
Statement executed successfully.

Query Details

Query duration 60ms

- Task **stops executing**
- stored Procedure **will not run**
- No new audit records inserted

Restart the Task :

```
426 ALTER TASK orders_audit_task RESUME;
427
```

Results | Chart

status
Statement executed successfully.

Query Details

Query duration 240ms

Rows 1

- Task resumes as per schedule
- Stored Procedure executes again
- Data loading continues normally

Simple Hierarchy :

Feature	Behavior
Dynamic Table failure	Goes to FAILED state
Data loss	✗ No
Fix SQL	Auto-recovers
Manual restart	✗ Not required
Task suspend	Stops execution
Task resume	Continues scheduling